

# Technical Documentation for a First-Order Delta-Sigma ADC

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## Abstract

This document describes a MATLAB/Simulink implementation of a basic high-resolution Delta-Sigma Analog-to-Digital Converter (ADC). The design uses a first-order, single-bit Delta-Sigma modulator followed by a third-order CIC decimation filter. It is evaluated for oversampling ratios of 64, 128, and 256 using a low-frequency sinusoidal input and a separate DC-input check, with Signal-to-Noise Ratio (SNR) and Effective Number of Bits (ENOB) used as the primary dynamic performance metrics.

## 1 Design Overview

The project models and simulates a simple Delta-Sigma ADC for high-precision conversion of low-frequency analog sensor signals. The conversion chain consists of:

- A first-order, single-bit Delta-Sigma modulator containing an integrator, comparator, and one-bit DAC feedback path.
- A digital decimation filter, implemented as a CIC filter, to suppress shaped quantization noise and reduce the sample rate.
- Performance evaluation using both sinusoidal and DC input conditions, including SNR and ENOB estimation across different oversampling ratios for the sinusoidal test case.

The project contains the following implementation artefacts:

- Simulink model: `InterIIT_ISRO_VLSI/DeltaSigma_ADC.slx`.
- MATLAB driver script: `InterIIT_ISRO_VLSI/Simulation_Code.txt`.
- Workspace file: `InterIIT_ISRO_VLSI/Matlab.mat`.
- LTspice analog schematic: `InterIIT_ISRO_VLSI/DeltaSigmaAnalogBreakdown.asc`.

## 2 Delta-Sigma ADC Architecture

The implemented ADC uses oversampling and noise shaping. The analog input is sampled at  $F_s = F_{out} \text{OSR}$ , passed through a first-order feedback modulator, and converted into a dense one-bit stream. A CIC (Cas-

caded Integrator–Comb) decimation filter then low-pass filters the bitstream and reduces the rate back to  $F_{out}$ .

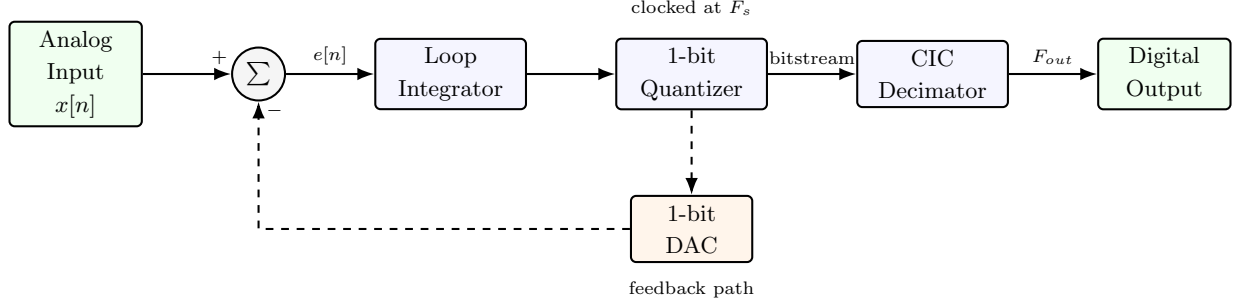


Figure 1: Functional architecture of the first-order Delta-Sigma ADC.

## 2.1 LTspice Analog Circuit Breakdown

The analog-level circuit was also captured through LTspice. LTspice is a **completely free to use** SPICE simulator for schematic capture, transient simulation, and waveform inspection. In this project, it provides a component-level cross-check of the same first-order Delta-Sigma loop before interpreting the Simulink and MATLAB results.

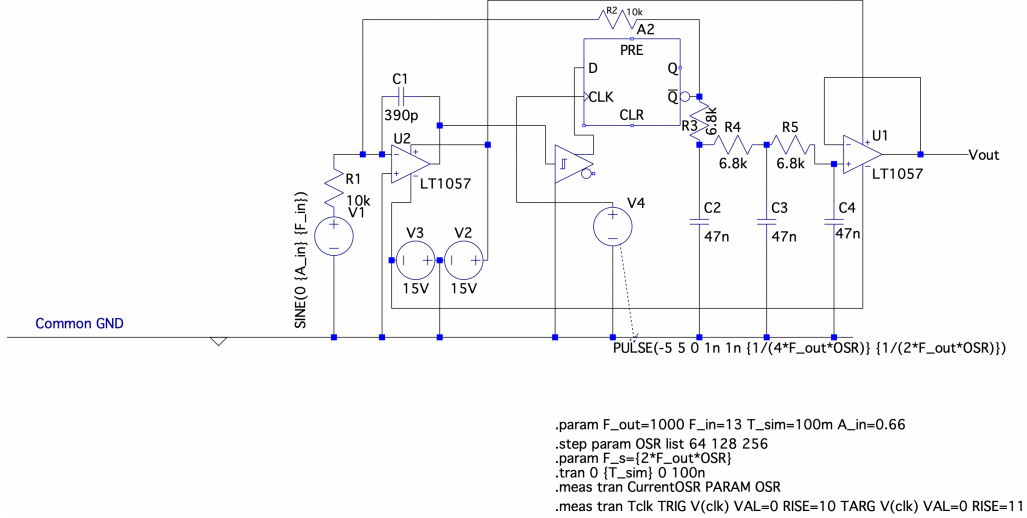


Figure 2: LTspice analog breakdown of the first-order Delta-Sigma converter.

Figure 2 shows the LT1057-based analog realization of the converter loop. The sinusoidal source uses  $A_{in} = 0.66$  and  $F_{in} = 13$  Hz, matching the low-frequency test condition used in the main simulations. The input resistor and feedback capacitor form the integrating stage, while the Schmitt/comparator and D flip-flop create the clocked one-bit decision path. The feedback branch closes the Delta-Sigma loop, and the three-section RC network using 6.8 kΩ resistors and 47 nF capacitors smooths the bitstream before the final LT1057 output buffer. The schematic also steps  $OSR = 64, 128, 256$ . The LTspice clock is  $F_{clk,LT} = 2F_{out}OSR$

because the analog pulse source is parameterized using half-period timing; the corresponding Delta-Sigma sample/update rate is therefore  $F_s = F_{clk,LT}/2 = F_{out}OSR$ . This maps the analog clock notation directly to the MATLAB/Simulink definition.

## 2.2 Theoretical Fundamentals

For a signal bandwidth  $B$ , a conventional Nyquist-rate ADC must sample at least at  $2B$  to avoid aliasing. A Delta-Sigma ADC instead samples much faster than this limit, with the oversampling ratio defined as

$$OSR = \frac{F_s}{2B}. \quad (1)$$

In this implementation, the modulator sampling rate is set as  $F_s = F_{out}OSR$  and the decimated output rate is  $F_{out} = 1000$  samples/s, so the useful output Nyquist band is below  $F_{out}/2 = 500$  Hz. The 13 Hz test tone therefore lies well inside the low-frequency signal band, while quantization noise is shaped toward higher frequencies before digital decimation.

## 2.3 First-Order Noise Shaping

The noise-shaping property of a first-order Delta-Sigma modulator can be analyzed using the additive quantization-noise model. Let  $q[n]$  denote the quantization error, and assume that the quantizer output can be represented as

$$y[n] = u[n] + q[n],$$

where  $u[n]$  is the integrator output. The modulator dynamics are

$$u[n] = u[n-1] + x[n] - y[n-1].$$

Taking the  $z$ -transform gives

$$(1 - z^{-1})U(z) = X(z) - z^{-1}Y(z).$$

Using

$$U(z) = Y(z) - Q(z),$$

and substituting into the previous equation,

$$(1 - z^{-1})(Y(z) - Q(z)) = X(z) - z^{-1}Y(z),$$

which simplifies to

$$\begin{aligned} Y(z) - Q(z) - z^{-1}Y(z) + z^{-1}Q(z) &= X(z) - z^{-1}Y(z), \\ Y(z) &= X(z) + (1 - z^{-1})Q(z). \end{aligned}$$

The key input-output relation is therefore

$$Y(z) = X(z) + (1 - z^{-1})Q(z). \quad (2)$$

Hence, the signal transfer function (STF) and noise transfer function (NTF) are

$$STF(z) = \left. \frac{Y(z)}{X(z)} \right|_{Q=0} = 1, \quad (3)$$

$$NTF(z) = \left. \frac{Y(z)}{Q(z)} \right|_{X=0} = 1 - z^{-1}. \quad (4)$$

Evaluating the NTF on the unit circle,

$$\begin{aligned} |\text{NTF}(e^{j\omega})| &= |1 - e^{-j\omega}| \\ &= \sqrt{(1 - \cos \omega)^2 + \sin^2 \omega} \\ &= \sqrt{2 - 2 \cos \omega}. \end{aligned}$$

Thus,

$$|\text{NTF}(e^{j\omega})| = 2 \sin\left(\frac{\omega}{2}\right). \quad (5)$$

Since

$$|\text{NTF}(e^{j\omega})| \rightarrow 0 \quad \text{as} \quad \omega \rightarrow 0,$$

the quantization noise is strongly attenuated at low frequencies and pushed toward higher frequencies. Therefore, a first-order Delta-Sigma modulator acts as a high-pass filter for quantization noise while passing the input signal with unity gain.

## 2.4 Modulator Model

The actual Simulink implementation uses a nonlinear 1-bit quantizer. Its discrete-time equations are

$$\begin{aligned} e[n] &= x[n] - y[n-1], \\ u[n] &= u[n-1] + e[n]. \end{aligned}$$

The quantizer output is

$$y[n] = \begin{cases} +1, & u[n] \geq 0, \\ -1, & u[n] < 0. \end{cases}$$

Here,  $x[n]$  is the sampled input,  $u[n]$  is the integrator state, and  $y[n]$  is the one-bit quantizer output. For analytical purposes, this nonlinear quantizer is replaced by the additive noise model introduced in the previous subsection, which leads to the STF and NTF expressions derived above.

## 2.5 CIC Decimation Filter

The decimation stage is implemented using the Simulink CIC Decimation block. The documented configuration is:

- Decimation factor:  $R = \text{OSR}$ .
- Differential delay:  $M = 1$ .
- Number of sections:  $N = 3$ .
- Internal arithmetic: full precision, with 32-bit output word length.

The CIC transfer function is

$$H(z) = \left( \frac{1 - z^{-RM}}{1 - z^{-1}} \right)^N. \quad (6)$$

For this model, the DC gain is  $R^N = \text{OSR}^3$ . The MATLAB script compensates this gain using

$$y_{\text{norm}}[n] = \frac{y_{\text{CIC}}[n]}{\text{OSR}^3}. \quad (7)$$

### 3 Simulation Configuration

The driver script evaluates the ADC for three oversampling ratios. Table 1 summarizes the key simulation settings.

Table 1: Simulation parameters used by the driver script.

| Parameter | Value            | Purpose                                             |
|-----------|------------------|-----------------------------------------------------|
| $F_{out}$ | 1000 samples/s   | Output data rate after decimation                   |
| $F_{in}$  | 13 Hz            | Low-frequency sinusoidal test signal                |
| $A_{in}$  | 0.66             | Input sine amplitude used by the script             |
| $T_{sim}$ | 2 s              | Intended simulation duration in the driver          |
| OSR       | 64, 128, 256     | Oversampling ratio sweep                            |
| $F_s$     | 64, 128, 256 kHz | Modulator sampling rate, $F_s = F_{out} \text{OSR}$ |
| CIC order | 3                | Noise suppression before decimation                 |
| CIC delay | 1                | Differential delay parameter                        |

In the simulation driver,  $A_{in} = 0.66$  before running each OSR case.

#### 3.1 Input Test Cases

Two input conditions are considered. The 13 Hz sine wave is the dynamic test case used for FFT-based SNR and ENOB estimation. A DC input is also included as a static conversion check. For a constant input level, the Delta-Sigma loop should produce a one-bit stream whose average density represents the applied DC value, and the normalized CIC output should settle near the corresponding constant output level.

The DC case is interpreted differently from the sine-wave case. Since a DC input appears at the zero-frequency bin, it is not suitable for the tone-bin SNR calculation used in Table 2. The mean-removal step in the FFT flow intentionally suppresses DC before dynamic SNR estimation. Therefore, the DC-input run is used to verify static tracking, polarity, and settling behavior, while the 13 Hz sine-wave run is used to compute the reported SNR and ENOB values.

### 4 Performance Evaluation Method

After each Simulink run, the MATLAB script performs the following post-processing steps:

1. Convert the saved one-bit stream and filtered output to double precision arrays.
  2. Normalize the CIC output by dividing by  $\text{OSR}^3$ .
  3. Discard the first 30% of the decimated output to remove startup transient effects.
  4. Remove the mean value to suppress DC offset before FFT analysis.
- ★ Use  $N_{FFT} = \min(F_{out}, \text{length}(y))$  samples for spectrum estimation. Here,  $N_{FFT}$  is the FFT length, meaning the number of decimated output samples used to compute the spectrum. This value sets the

FFT record length and the frequency-bin spacing,  $\Delta f = F_{out}/N_{FFT}$ . Since  $F_{out} = 1000$  samples/s, using  $N_{FFT} = F_{out}$  means the FFT looks at one second of the final decimated waveform. A one-second record gives simple 1 Hz frequency bins, so the 13 Hz input tone falls exactly on the 13 Hz FFT bin instead of between two bins. The `length(y)` term is a safety check: after removing the startup transient, the remaining output may sometimes contain fewer than 1000 samples, so the script uses only the samples that actually exist. Taking the minimum of these two values therefore prevents indexing errors, keeps the analysis stable for every OSR case, and gives consistent spectral resolution whenever a full one-second record is available.

5. Estimate signal power around the 13 Hz tone using the tone bin and its adjacent bins.
6. Estimate noise power from the remaining positive-frequency bins, excluding DC and the signal bins.

The reported SNR and ENOB are computed as

$$\text{SNR}_{dB} = 10 \log_{10} \left( \frac{P_{signal}}{P_{noise}} \right), \quad (8)$$

$$\text{ENOB} = \frac{\text{SNR}_{dB} - 1.76}{6.02}. \quad (9)$$

## 5 Representative Results

Table 2 gives results obtained directly from MATLAB post-processing of the Simulink output. The trend is the key design result: increasing OSR reduces in-band quantization noise and improves ENOB.

Table 2: Representative SNR and ENOB trend for the configured ADC model.

| OSR | Modulator rate $F_s$ | SNR      | ENOB       |
|-----|----------------------|----------|------------|
| 64  | 64 kHz               | 56.07 dB | 9.02 bits  |
| 128 | 128 kHz              | 65.17 dB | 10.53 bits |
| 256 | 256 kHz              | 74.00 dB | 12.00 bits |

The target ENOB range of approximately 10–12 bits is reached by the  $\text{OSR} = 128$  and  $\text{OSR} = 256$  cases, which produce 10.53 bits and 12.00 bits, respectively. The  $\text{OSR} = 64$  case produces 9.02 bits, so it should be treated as a lower-resolution baseline. This limitation is expected for a simple first-order, single-bit Delta–Sigma architecture because the lowest OSR leaves more shaped quantization noise inside the signal band. The result confirms the usual Delta–Sigma trade-off that higher oversampling improves resolution at the cost of a higher modulator clock and greater digital filtering activity.

## 6 Generated Outputs

The MATLAB script generates the following diagnostic outputs for each OSR value:

- A stair plot of the first samples of the one-bit Delta–Sigma bitstream.
- A time-domain plot of the reconstructed and normalized output waveform.

- A normalized FFT magnitude plot of the filtered output.
- A semilog plot of SNR versus OSR.
- A semilog plot of ENOB versus OSR.
- A printed table containing OSR, SNR in dB, and ENOB in bits.
- DC-input bitstream and reconstructed-output plots for the  $\text{OSR} = 64$  static conversion check.

These outputs provide direct visibility into the bitstream behavior, filtered digital output, spectral characteristics, and performance metrics of the ADC chain.

## 6.1 Simulation Output

The figures below are ordered by increasing oversampling ratio. For each OSR case, the one-bit bitstream is shown first, followed by the reconstructed output waveform and the corresponding output FFT; the final figure summarizes SNR and ENOB across the OSR sweep.

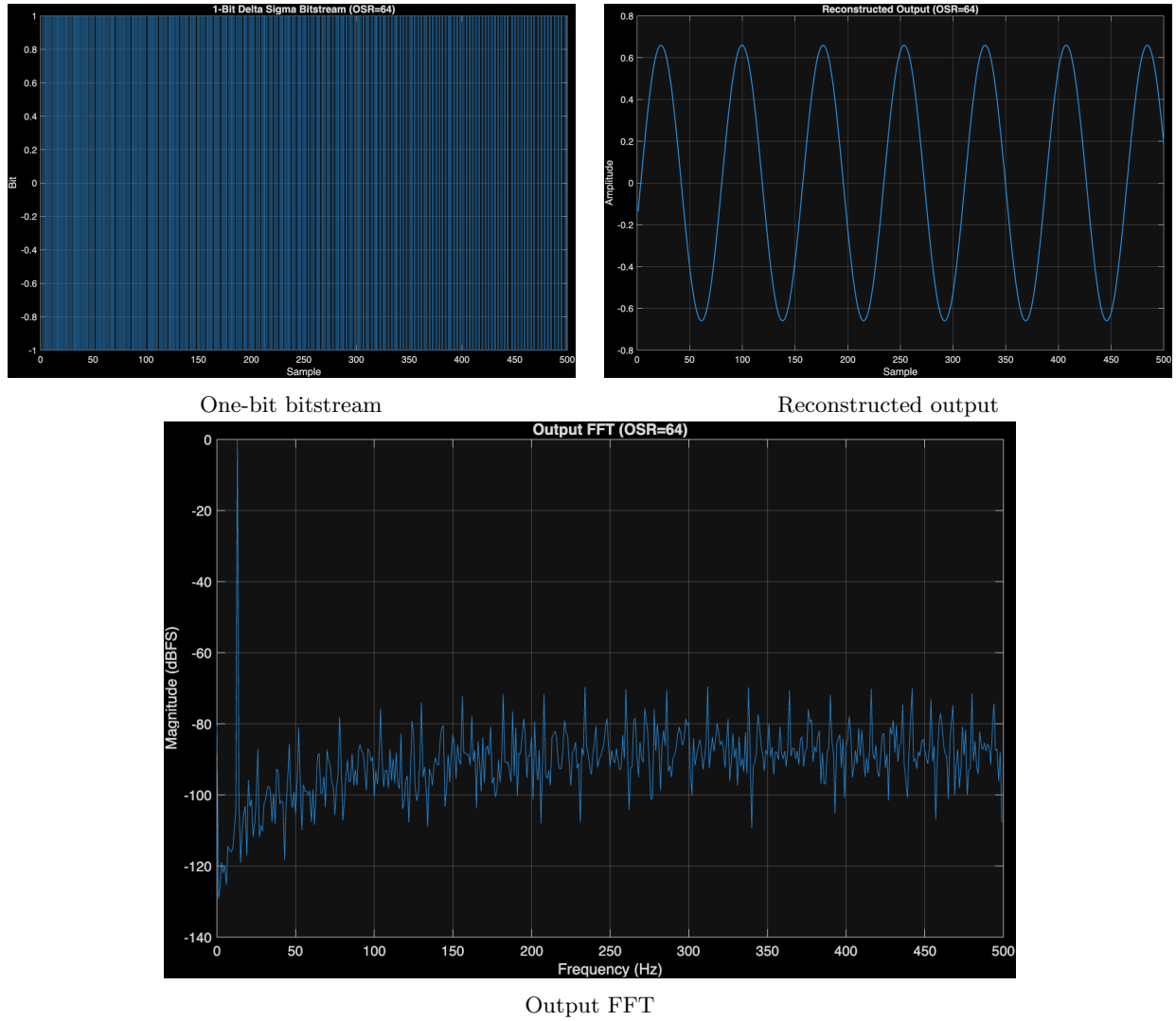
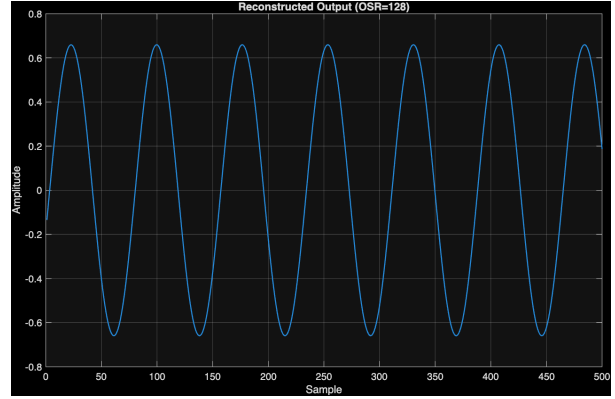
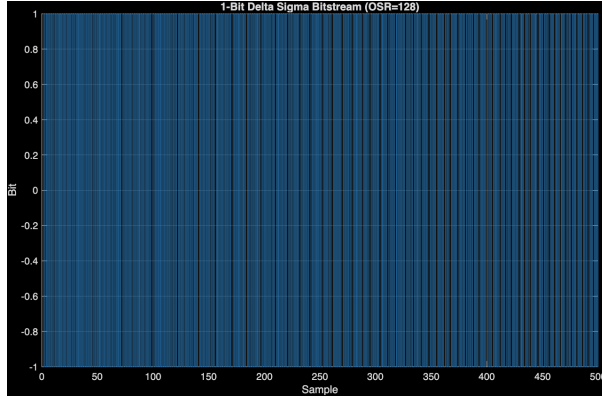


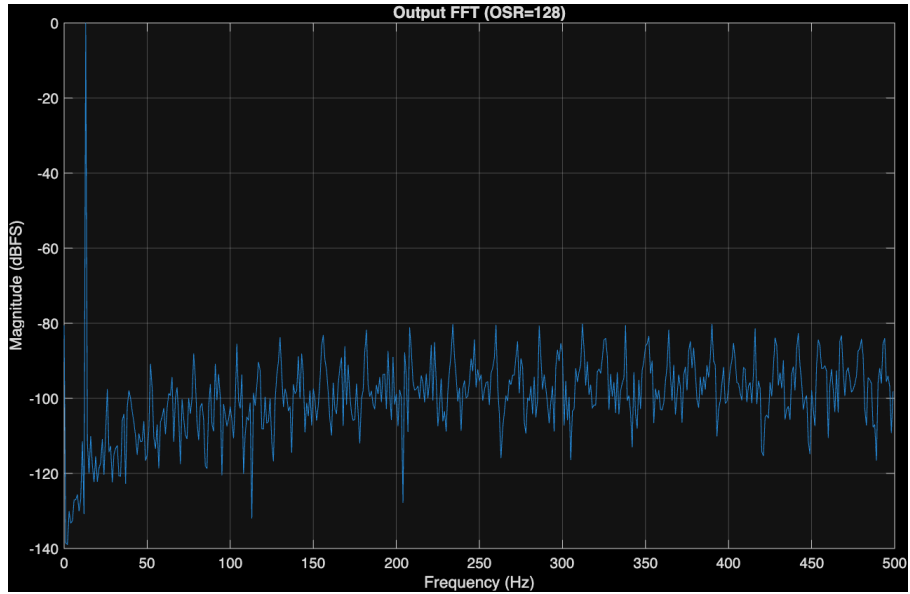
Figure 3: Diagnostic simulation outputs for OSR = 64.





One-bit bitstream

Reconstructed output



Output FFT

Figure 4: Diagnostic simulation outputs for  $\text{OSR} = 128$ .

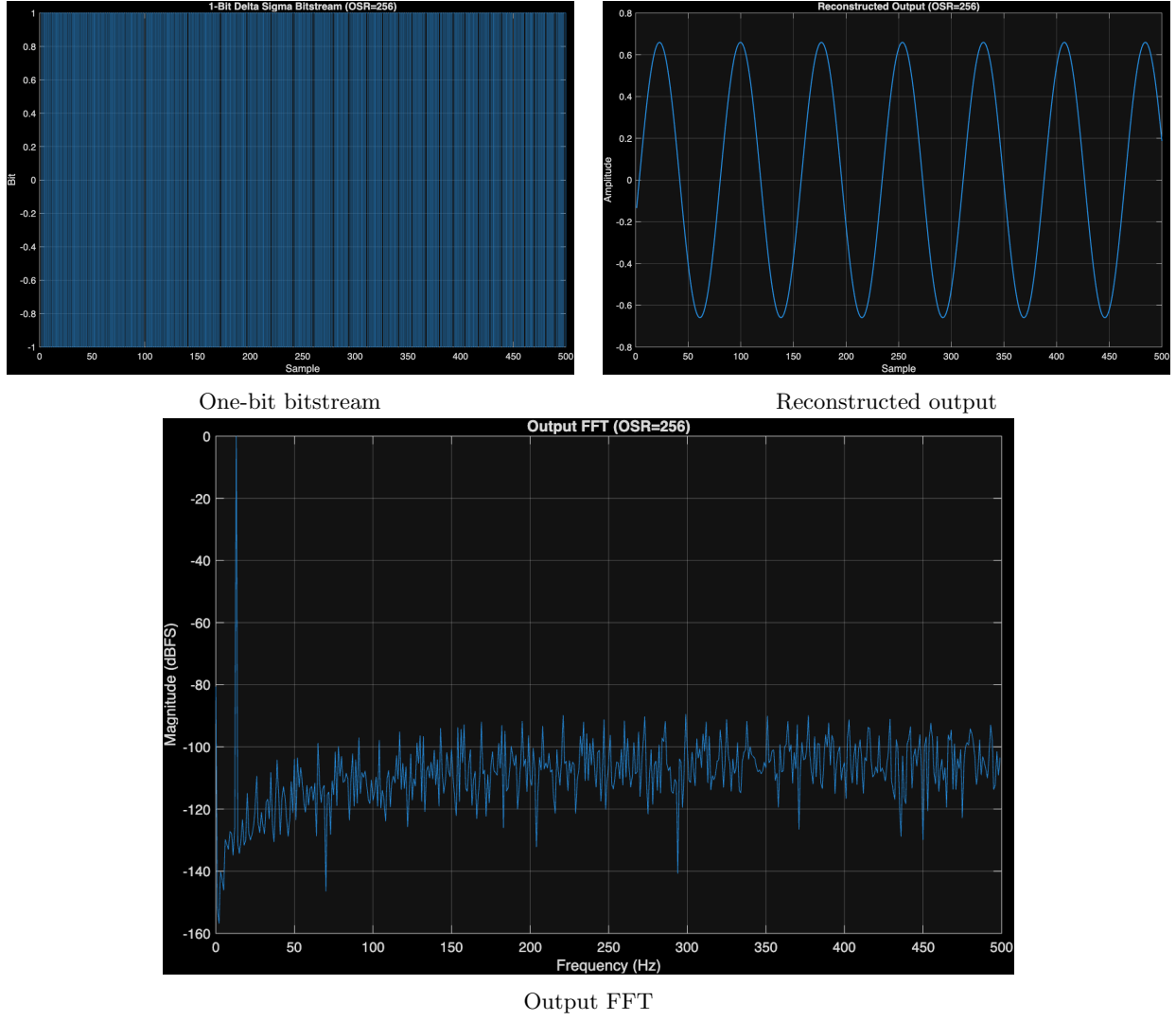


Figure 5: Diagnostic simulation outputs for OSR = 256.

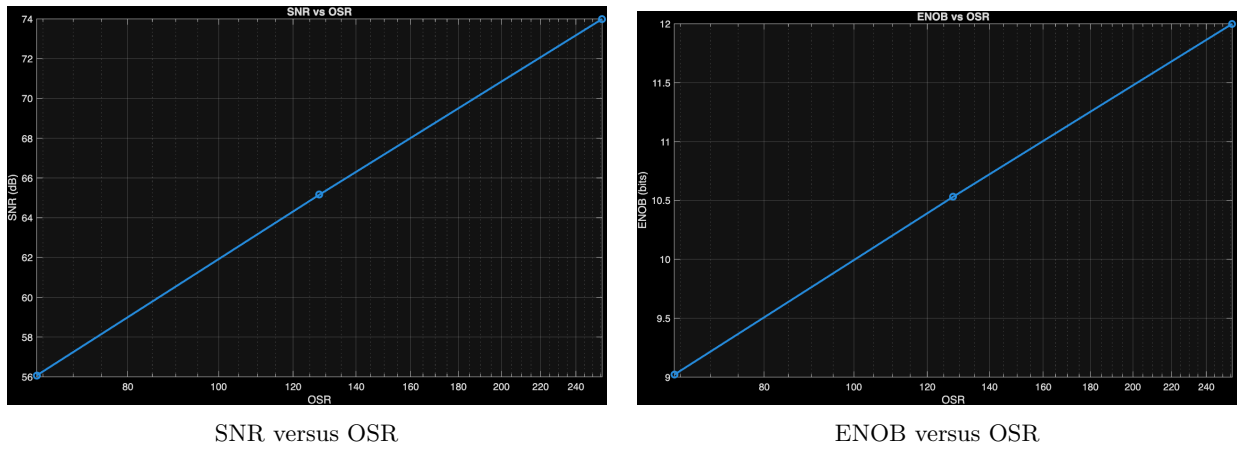


Figure 6: Summary performance trends across the OSR sweep.

## 6.2 DC Input Static Check

Figure 7 shows the DC-input static check for  $\text{OSR} = 64$ . The DC simulation was performed for all specified OSR values, but only the  $\text{OSR} = 64$  case is documented here as a representative example. The one-bit stream continues switching rapidly, but its average density represents the applied DC level. After CIC decimation and gain normalization, the reconstructed output settles near the constant input level, confirming static tracking for the baseline OSR case.

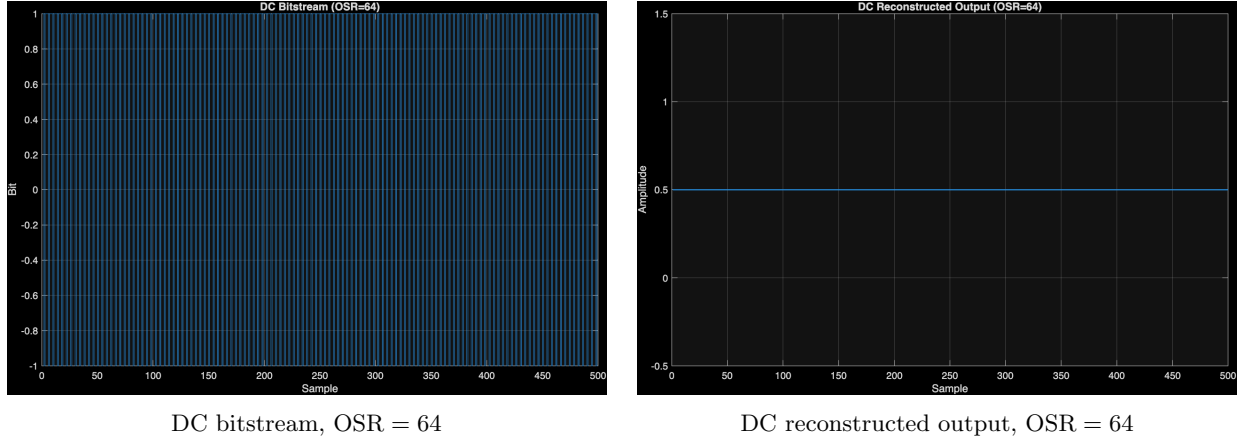


Figure 7: DC-input static-check outputs for  $\text{OSR} = 64$ .

## 7 Design Coverage Summary

Table 3: Summary of the implemented design and analysis coverage.

| Design aspect                     | Documentation coverage                                                                                                                      |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Literature study and fundamentals | The design uses oversampling, first-order noise shaping, single-bit quantization, decimation concepts, and the z-domain STF/NTF derivation. |
| First-order modulator design      | Implemented in Simulink using integrator, relay comparator, and feedback subtraction.                                                       |
| Digital decimation filter         | Implemented as a third-order CIC decimator with $R = \text{OSR}$ .                                                                          |
| Performance analysis              | SNR, ENOB, waveform, FFT plots, the $\text{OSR} = 64$ DC-input static check, and the FFT windowing assumption are documented.               |
| Technical documentation           | This document organizes the design architecture, simulation flow, generated results, limitations, and possible extensions.                  |

## 8 Limitations and Possible Extensions

The model intentionally focuses on a simplified first-order, single-bit ADC architecture. The following improvements may be considered for future work:

- At  $\text{OSR} = 64$ , the measured ENOB is about 9.02 bits, below the 10–12 bit target. This is an expected limitation of the simple first-order, single-bit architecture, while the  $\text{OSR} = 128$  and  $\text{OSR} = 256$  cases meet the target range.
- Compare the current CIC filter against Sinc or FIR compensation filters to reduce passband droop.
- Implement the CIC decimation chain in Verilog or VHDL for hardware-oriented evaluation.
- Evaluate second-order or higher-order Delta–Sigma modulators for improved in-band noise shaping.
- Study multi-bit quantization to improve SNR while considering DAC linearity requirements.
- Simulate offset and low-frequency noise mitigation methods such as chopping or auto-zeroing.

## 9 Conclusion

The documented design implements a first-order, single-bit Delta–Sigma modulator, integrates a third-order CIC decimation filter, and evaluates the complete ADC chain over an OSR range of 64 to 256. The documented performance trend shows that increasing OSR from 64 to 256 improves ENOB from roughly 9 bits to roughly 12 bits, demonstrating the expected advantage of oversampling and noise shaping for high-resolution low-frequency sensor conversion.